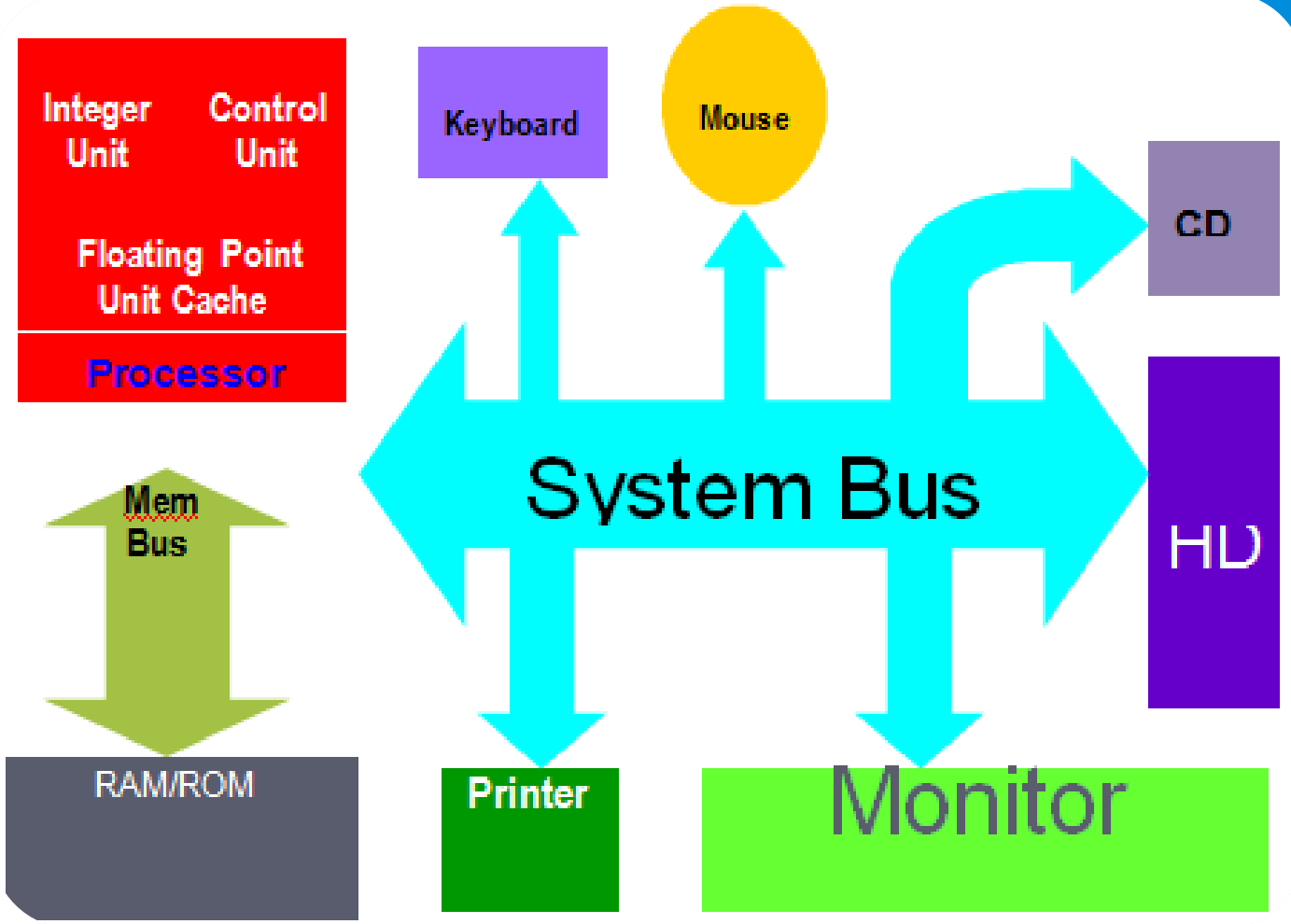
Operating Systems

Kernel and Process Management

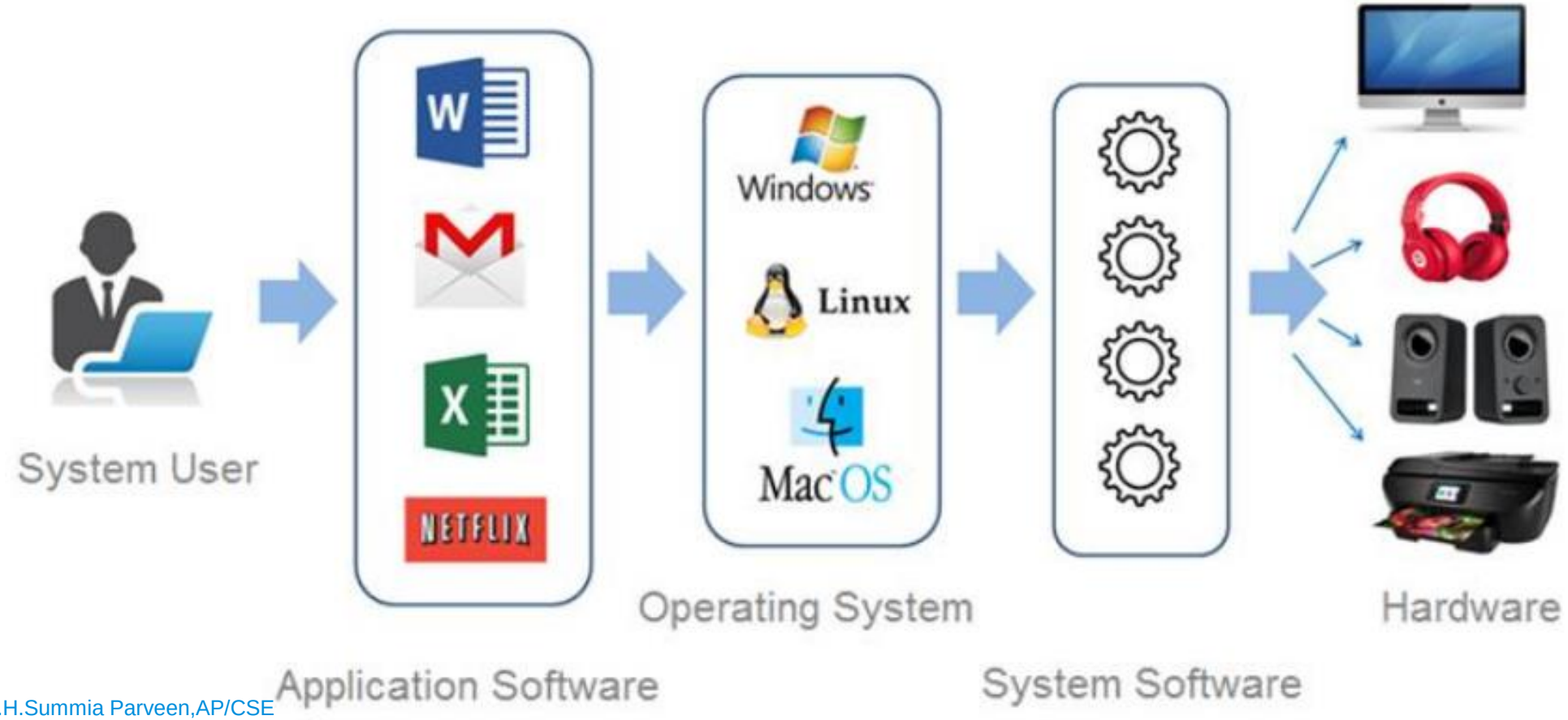
UNIT I

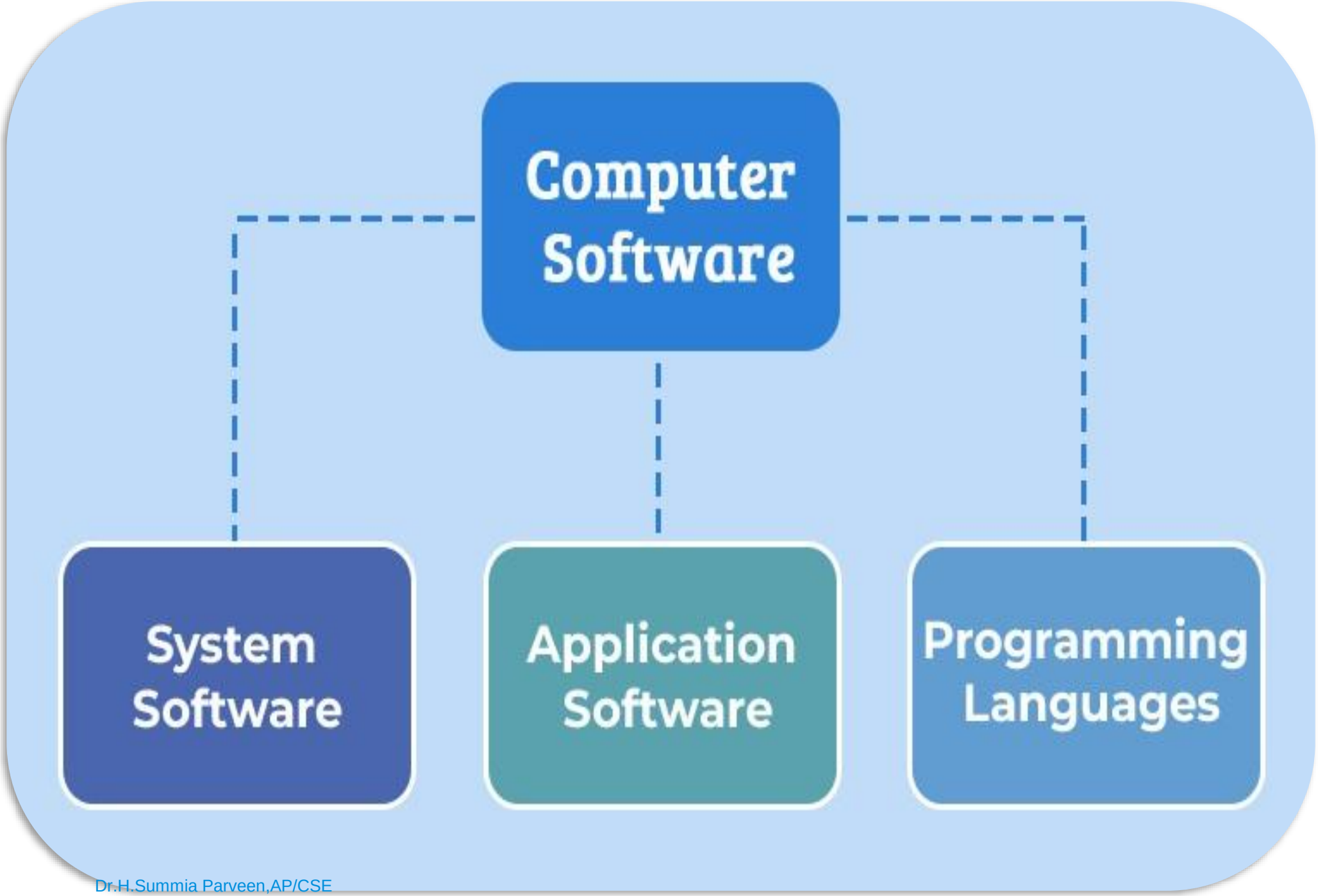
OS X
Windows
Linux
Xen
Red Hat
Fedora
CentOS
Debian
Apple
Ubuntu
Mint
SUSE
Mageia
Arch Linux
Slackware
Mandriva
Gentoo
FreeBSD
OpenBSD
NetBSD
NetBSD
DragonFly BSD
Darwin

ORGANIZATION OF A COMPUTER



HARDWARE
SOFTWARE
Java
Windows 7
McAfee


System User
W
NETFLIX
Windows
Linux
MacOS
Hardware
Operating System
Application Software
System Software


Computer Software
System Software
Application Software
Programming Languages

SYSTEM SOFTWARE

- The system software is a type of computer software that is designed for running the computer hardware parts and the application programs. It is the platform provided to the computer system where other computer programs can execute. The system software act as a middle layer between the user applications and hardware.

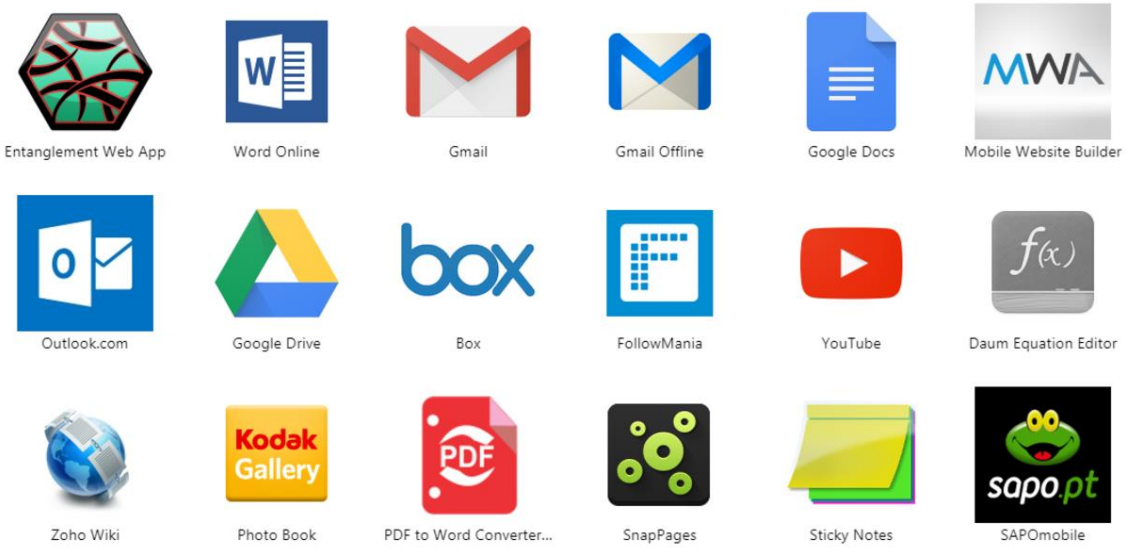
- **Operating System**
- **Device Drivers**
  - *Mouse Driver*
  - *Print Driver*
- **Language Processors**
- **Firmware**

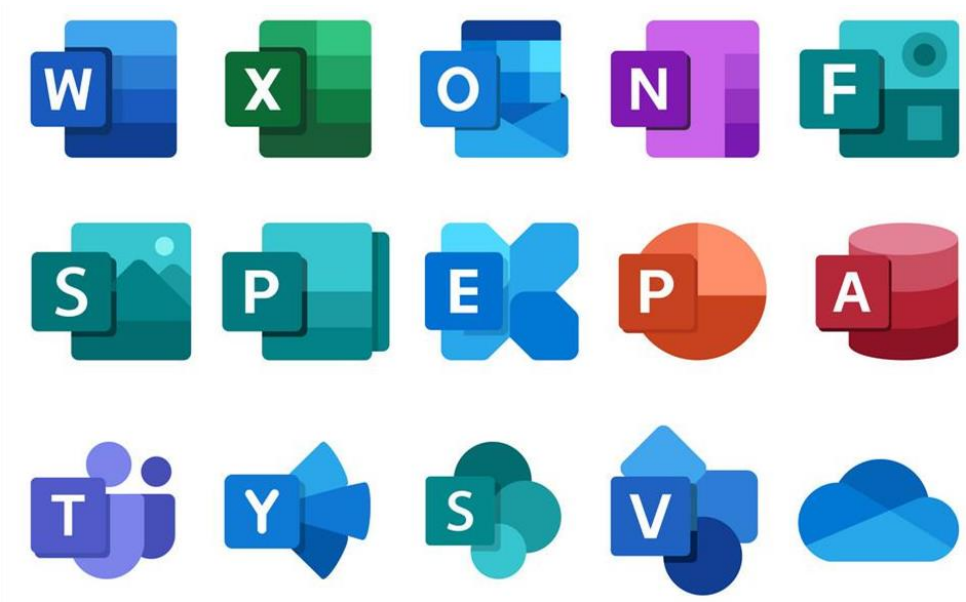
SYSTEM SOFTWARE

Android Studio, Xcode, Notepad++, Visual Studio Code, Eclipse, Github, Gitlab, Atom

APPLICATION SOFTWARE

- Software or program that is designed to complete a specific task





APPLICATION SOFTWARE

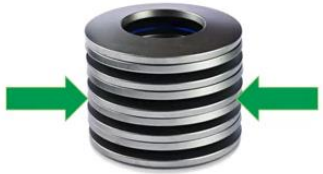
Need some environment to run and perform its tasks

UTILITY SOFTWARE

Utility software is a category of software that focuses on maintaining smooth working of the computing device along with the operating system.

ANTIVIRUS SOFTWARE

BACKUP SOFTWARE



DISK COMPRESSION TOOL

FILE MANAGER

What is a DEVICE driver?

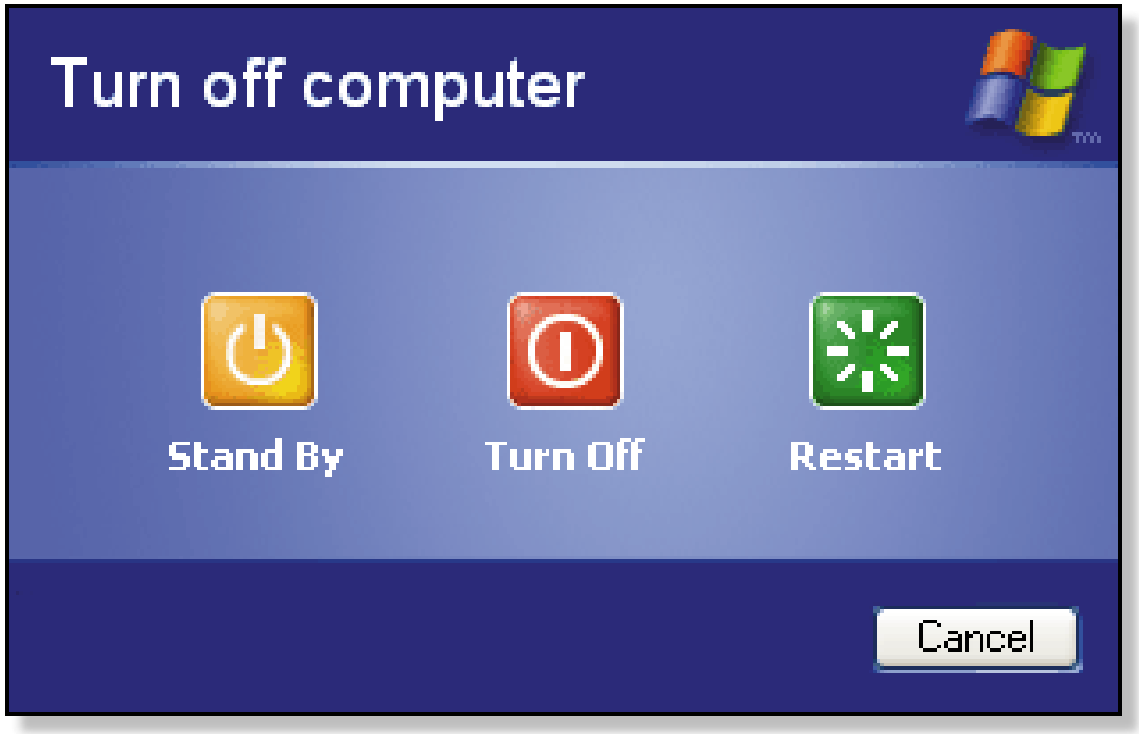
Program that tells operating system how to communicate with device

Driver

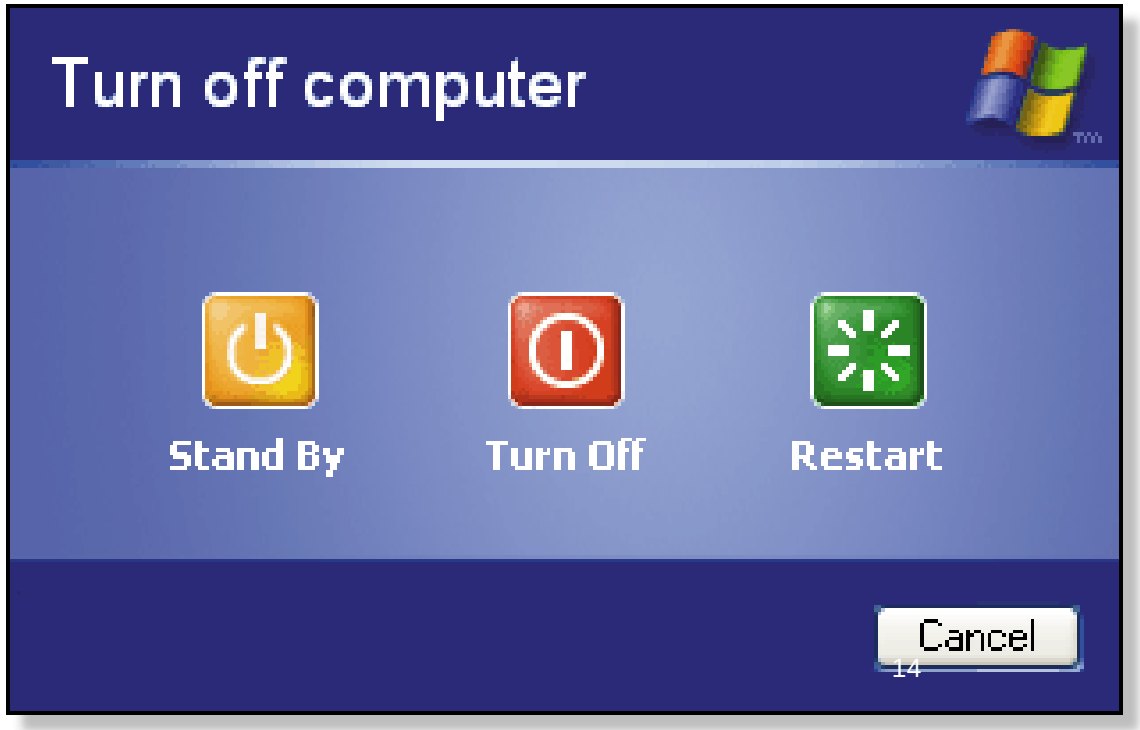
With Plug and Play, operating system automatically configures new devices as you install them

What is Booting?

- **Process of starting or restarting a computer**
  - **Cold boot** Turning on computer that has been powered off
  - **Warm boot** Restarting computer that is powered on



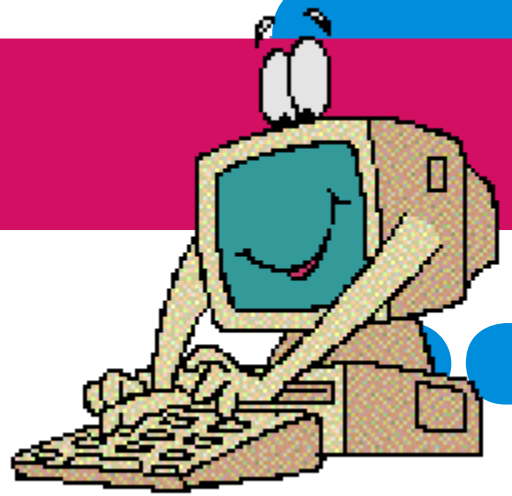
- **Process of starting or restarting a computer**
  - **Cold boot** Turning on computer that has been powered off
  - **Warm boot** Restarting computer that is powered on



What is an Operating System?

Dr.H.Summia Parveen,AP/CSE

- An Operating System is a system software that manages computer hardware and software resources and provides common services for computer programs.
- Every computer system must have at least one operating system to run other programs.
- The OS helps you to communicate with the computer without knowing how to speak the computer's language.
- It is not possible for the user to use any computer or mobile device without having an operating system.



Maid
(Kernel)

Homemaker
(Operating System)

Who manages everything at hom

OPERATING SYSTEM

OBJECTIVES

1.Convenience:

2.Efficiency:

3.Ability to evolve: